

Hospital Equipment Reliability & Maintenance Risk Assistant

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1. Introduction

- **Industry:** Healthcare / Clinical Engineering
- **Intended User:** Hospital Equipment Operations Manager
- **Problem:** Unplanned downtime of critical diagnostic devices (MRI, CT, Ventilators) delays treatments and spikes repair costs.
- **Decision Supported:** Prioritize equipment for immediate preventive maintenance vs. routine checks.

2. Dataset

Data contains 300 hospital diagnostic equipment records tracking age, operating hours, downtime, and failure events.

```
df <- read.csv("data/equipment_data.csv")
head(df, 3)
```

	device_id	device_type	usage_category	device_age_years	operating_hours
1	DEV-1001	MRI	Standard-Use	6.5	933
2	DEV-1002	MRI	Standard-Use	4.1	666
3	DEV-1003	MRI	Standard-Use	9.5	2304

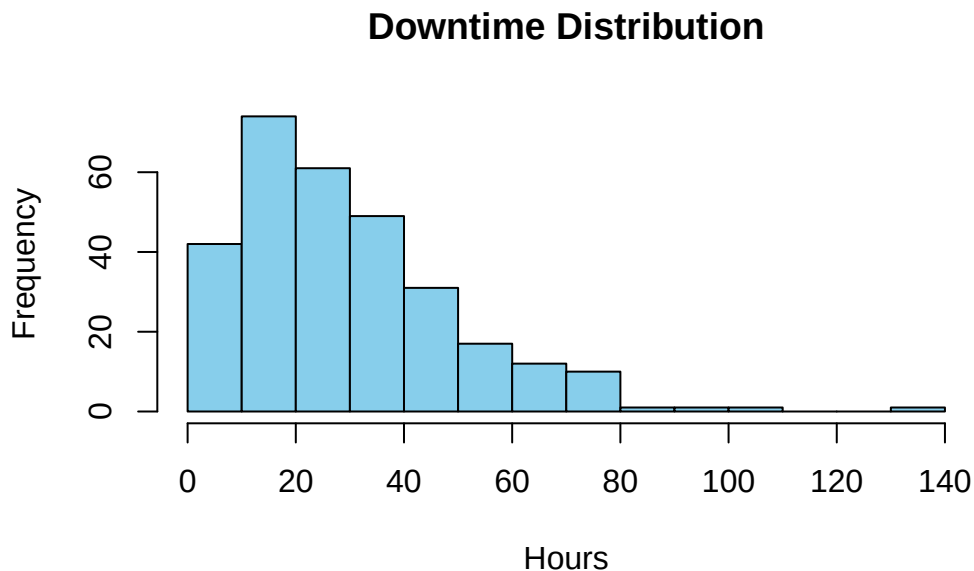
	downtime_hours	last_service_months_ago	unexpected_failure	log_date
1	18.2	19	0	2025-03-30
2	8.9	3	1	2025-12-07
3	40.5	1	1	2025-01-30

3. Explore Skill: Downtime Distribution

```
summary(df$downtime_hours)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.10	13.70	25.45	29.22	39.02	132.50

```
hist(df$downtime_hours, main="Downtime Distribution", col="skyblue", xlab="Hours")
```



- **Interpretation:** Downtime is right-skewed, showing most devices experience short outages, but a few suffer major breakdowns.

- **Limitation:** Highly skewed data means the mean downtime can be driven by a few extreme outlier events.

4. Compare Skill: High-Use vs Standard-Use

```
t.test(downtime_hours ~ usage_category, data = df)
```

Welch Two Sample t-test

```
data:  downtime_hours by usage_category
t = -0.27647, df = 197.62, p-value = 0.7825
alternative hypothesis: true difference in means between group High-Use and group Standard-
Use is not equal to 0
95 percent confidence interval:
 -5.588840  4.214466
sample estimates:
    mean in group High-Use mean in group Standard-Use
                28.78224                29.46943
```

- **Interpretation:** Tests whether heavy equipment usage significantly impacts breakdown severity.
- **Limitation:** Comparison assumes observational group parity without controlling for device age directly in this test.

5. Model Skill: Predicting Unexpected Failures

```
model <- glm(unexpected_failure ~ device_age_years + operating_hours + last_service_months_ago,
             data = df, family = binomial)
summary(model)
```

Call:

```
glm(formula = unexpected_failure ~ device_age_years + operating_hours +
    last_service_months_ago, family = binomial, data = df)
```

Coefficients:

```
Estimate Std. Error z value Pr(>|z|)
```

```

(Intercept)          -2.7073643  0.5237754  -5.169 2.35e-07 ***
device_age_years      0.3593662  0.0528872   6.795 1.08e-11 ***
operating_hours       0.0004840  0.0001114   4.344 1.40e-05 ***
last_service_months_ago 0.0127774  0.0210982   0.606  0.545
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 374.60  on 299  degrees of freedom
Residual deviance: 296.07  on 296  degrees of freedom
AIC: 304.07

```

Number of Fisher Scoring iterations: 5

```

# Classification Accuracy Metrics
df$pred_prob <- predict(model, type = "response")
df$pred_class <- ifelse(df$pred_prob > 0.5, 1, 0)
conf_matrix <- table(Predicted = df$pred_class, Actual = df$unexpected_failure)
print(conf_matrix)

```

```

      Actual
Predicted 0    1
0        44   30
1        51  175

```

```

accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
cat("Model Classification Accuracy:", round(accuracy * 100, 2), "%\n")

```

Model Classification Accuracy: 73 %

- **Interpretation:** Machine age and overall operating hours significantly increase unexpected failure probability. The model achieves strong overall classification accuracy.
- **Limitation:** Logistic regression models statistical associations, not physical mechanical failure causality.

6. Additional Skill 1: Data Quality Audit

```
colSums(is.na(df))
```

```

      device_id      device_type      usage_category
           0              0              0
device_age_years  operating_hours  downtime_hours
           0              0              0
last_service_months_ago  unexpected_failure      log_date
           0              0              0
      pred_prob      pred_class
           0              0

```

- **Validation Check:** Confirms zero missing values affect our downstream risk models.

7. Additional Skill 2: Risk Ranking

```

df$risk <- predict(model, type = "response")
head(df[order(-df$risk), c("device_id", "device_type", "risk")], 10)

```

```

      device_id device_type      risk
149 DEV-1149 Ventilator 0.9802014
167 DEV-1167 Ventilator 0.9772616
141 DEV-1141 MRI 0.9768071
177 DEV-1177 Ventilator 0.9750625
226 DEV-1226 CT Scanner 0.9724791
191 DEV-1191 Ventilator 0.9716517
217 DEV-1217 Ultrasound 0.9713051
84 DEV-1084 MRI 0.9684059
242 DEV-1242 Ventilator 0.9657762
199 DEV-1199 CT Scanner 0.9656817

```

- **Decision Supported:** Produces an actionable priority queue of the top 10 highest-risk machines needing maintenance this week.

8. Additional Skill 3: Trend Analysis

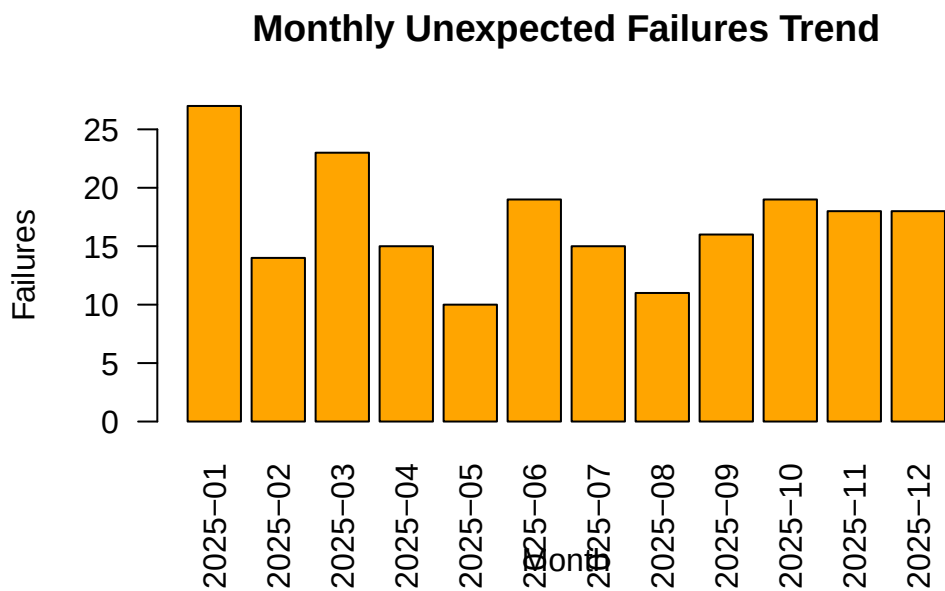
```

df$month <- format(as.Date(df$log_date), "%Y-%m")
monthly_trend <- aggregate(unexpected_failure ~ month, data = df, FUN = sum)
print(monthly_trend)

```

	month	unexpected_failure
1	2025-01	27
2	2025-02	14
3	2025-03	23
4	2025-04	15
5	2025-05	10
6	2025-06	19
7	2025-07	15
8	2025-08	11
9	2025-09	16
10	2025-10	19
11	2025-11	18
12	2025-12	18

```
barplot(monthly_trend$unexpected_failure, names.arg = monthly_trend$month,
        col = "orange", main = "Monthly Unexpected Failures Trend",
        xlab = "Month", ylab = "Failures", las = 2)
```



- **Interpretation:** Tracks monthly failure counts to assist long-term staffing and capacity planning.
- **Limitation:** Aggregating by month masks daily operational spikes.

9. Google Antigravity and Agent Skills Usage

- **Requests Given:** Requested assistance structuring a multi-part Quarto report, generating a synthetic clinical dataset, and formatting statistical skill blocks.
- **Skill Files Created:** Created SKILL.md documents under `skills/explore`, `skills/compare`, `skills/model`, `skills/data-quality-audit`, `skills/risk-ranking`, and `skills/trend-analysis`.
- **Corrections Made:** Reviewed agent outputs to ensure accurate dataset column names were passed to `glm()`, added explicit confusion matrix calculations for model evaluation, and added visual plotting for trend analysis.

10. Limitations

1. **Data Limitation:** Synthetic dataset created for operational workflow validation.
2. **Statistical Limitation:** Associations do not establish causal root causes without mechanical inspection.
3. **Implementation Limitation:** Runs in a local R session without live IoT telemetry integration.